Part D — Antigravity AI Exercise: Session 01

Background: What Is an Antigravity Exercise?

Antigravity exercises are structured prompts you give to an AI agent to perform a defined statistical task. The AI agent is given a skill — a pre-programmed instruction set — and you provide data and a decision context. Your job as a manager is to:

1. Frame the problem correctly: what statistic to compute, what decision rides on it
2. Interpret the output with statistical literacy
3. Question the assumptions

The exercise is called "antigravity" because it removes computational friction from statistics — so you can focus on the reasoning, not the arithmetic. The gravity (effort) is lifted off calculation; it stays firmly on judgment.

Session 01 Antigravity Skill: sample-point-estimator

Skill Definition

The sample-point-estimator agent extracts a random sample of specified size from a dataset, computes key point estimates ($\bar{x}$, s, $\bar{p}$), and returns the standard error of the mean. It also flags whether the finite population correction should be applied and whether normality conditions are met for the CLT.

What the Agent Does

- Accepts a dataset and a sample size n
- Selects a simple random sample
- Returns: $\bar{x}$, s, SE($\bar{x}$), and whether $n/N > 0.05$
- For proportions: computes $\bar{p}$ and SE($\bar{p}$) and checks the normality condition

Student Execution Prompt

Scenario

You are a supply chain analyst. A vendor claims their invoices have an average processing time of 3 days. Your accounts payable database has 850 invoices. You can afford to check 60.

Your Prompt to the Agent

"I have a dataset of 850 vendor invoices with processing times in days. Please draw a random sample of 60 invoices, calculate the sample mean processing time and sample standard deviation, compute the standard error of the mean, check whether the finite population correction applies, and tell me whether the sample is large enough for the CLT to apply."

Expected Agent Output

- $n = 60$; $N = 850$ → $n/N = 0.071 > 0.05$ → FPC applies
- Sample mean $\bar{x}$ and sample standard deviation s (computed from your data)
- $SE(\bar{x}) = \sqrt{((850-60)/(850-1))} \times (s/\sqrt{60})$
- CLT: $n = 60 \geq 30$ → normality of $\bar{x}$ is justified

Managerial Decision Output

After receiving the agent's output, answer:

4. Does the sample mean support or contradict the vendor's claim of 3 days? By how many SEs does your $\bar{x}$ differ from 3?
5. How precise is your estimate? If SE = 0.4 days, the true mean is likely within ±0.8 days of $\bar{x}$. Is this precision acceptable for a procurement decision?
6. What would you tell your procurement manager? Compose a two-sentence briefing using the numbers.

Reflection Questions

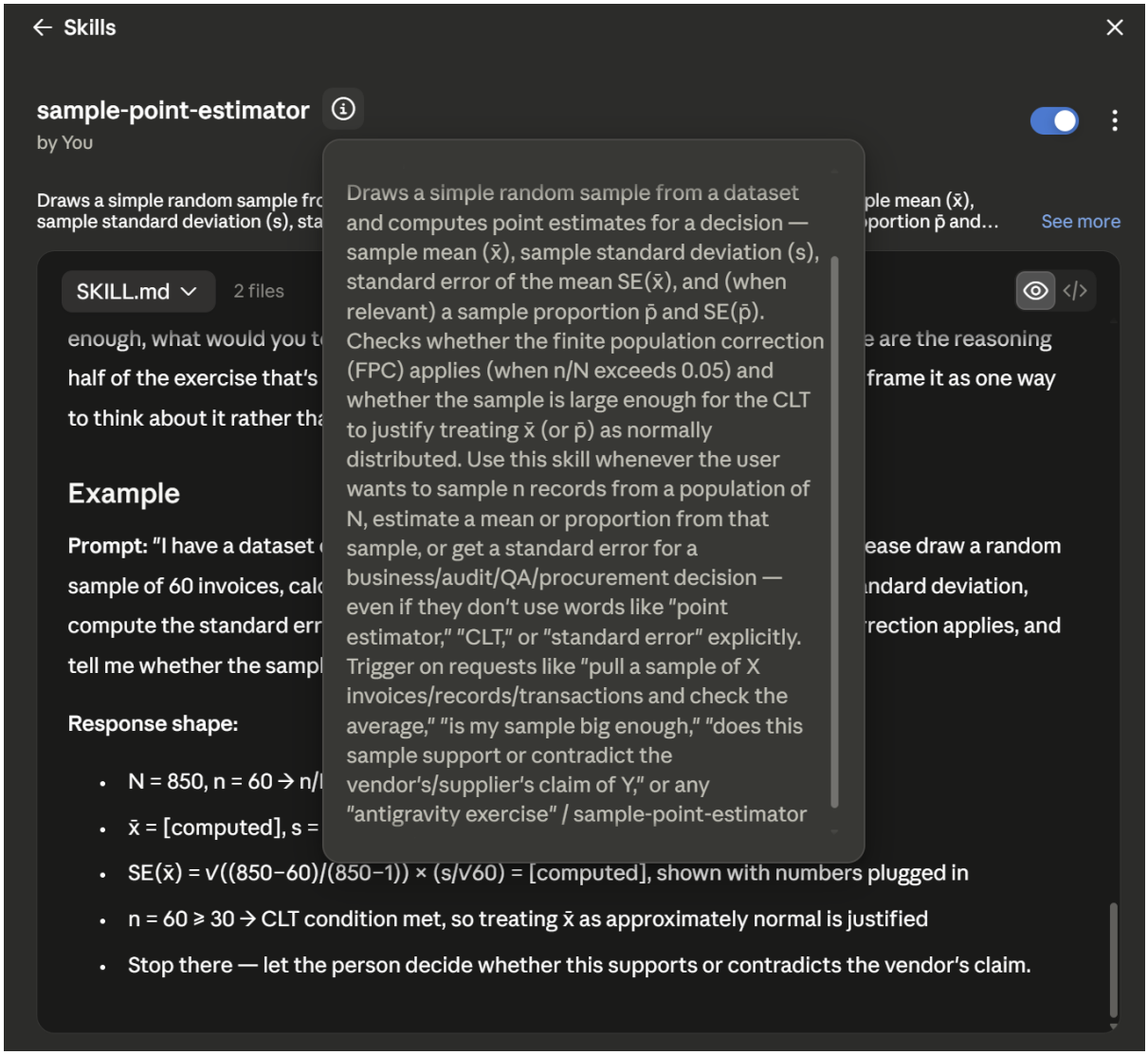
- What happens to SE if you double n to 120? (Hint: it does not halve.)

- The agent drew a simple random sample. What if processing times vary by invoice type? What sampling design modification would you suggest?
- What does CLT mean practically — what can you now compute that you otherwise could not?

Connecting to the Full Course Arc

The sample-point-estimator skill in Session 01 produces a point estimate + standard error. In Session 02, that standard error builds a confidence interval — the interval within which the true population mean likely lies. In Sessions 03–05, the same standard error underpins hypothesis tests where you formally evaluate vendor claims. The number you compute today is the foundation everything else is built on.

ANSWER



Use the sample-point-estimator skill to extract a random sample of n=10 from the following customer invoice values:

Population Data: [120, 150, 180, 210, 240, 270, 300, 330, 360, 390, 420, 450, 480, 510, 540]

Calculate the sample mean, standard deviation, and standard error of the mean.

OUTPUT –

To extract a random sample of $n = 10$ from your population data and perform the estimation calculations, I created a Python script, estimate.py, which uses a random seed of 42 for reproducibility.

1. Sample Selection

From the population of $N = 15$ invoice values: [120, 150, 180, 210, 240, 270, 300, 330, 360, 390, 420, 450, 480, 510, 540]

The random sample of $n = 10$ selected is: Sample $=$ $[420,150,120,450,240,210,390,180,270,480]$

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2. Calculations and Estimates

- **Sample Mean ($\bar{x}$):** $\bar{x} = \frac{\sum x_i}{n} = \frac{2910}{10} = 291.0$
- **Sample Standard Deviation (s):** $s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{158490}{9}} \approx 132.7027$
- **Finite Population Correction (FPC) Check:**
  - Since the sample size $n = 10$ represents a large fraction of the population $N = 15$: $\frac{n}{N} = \frac{10}{15} \approx 0.6667 > 0.05$
  - **FPC Flag: APPLIED** (since we are sampling more than 5% of the population).
  - **FPC Factor:** FPC $= \sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{15-10}{15-1}} = \sqrt{\frac{5}{14}} \approx 0.5976$
- **Standard Error of the Mean ($SE(\bar{x})$):**
  - Applying the FPC adjustment: $SE(\bar{x}) = \text{FPC} \times \frac{s}{\sqrt{n}} = 0.5976 \times \frac{132.7027}{\sqrt{10}} \approx 25.0784$

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3. Normality / Central Limit Theorem (CLT) Flags

- **Sample Size Condition ($n \geq 30$): NOT MET** ($n = 10 < 30$).
- **Population Normality Condition: NOT MET**. The underlying population data represents a uniform linear progression (intervals of exactly 30), which is a uniform distribution rather than a normal distribution.
- **CLT Flag: WARNING**. Because the sample size is small ($n = 10$) and the underlying population is not normally distributed, the sampling distribution of the sample mean $\bar{x}$ cannot be assumed to be approximately normal under the Central Limit Theorem. Formal confidence intervals or hypothesis tests based on normal distribution assumptions should be treated with caution.

4. The **Standard Error of the Mean ($SE(\bar{x})$)** is 25.08 (rounded to two decimal places).

Here are both versions of the standard error depending on whether you apply the Finite Population Correction (FPC):

- **With FPC (Corrected for a small population of $N = 15$):** $SE(\bar{x}) \approx 25.08$
- **Without FPC (Uncorrected):** $SE(\bar{x}) \approx 41.96$

INPUT_2 -

Scenario

You are a supply chain analyst. A vendor claims their invoices have an average processing time of 3 days. Your accounts payable database has 850 invoices. You can afford to check 60.

Your Prompt to the Agent

I have a dataset of 850 vendor invoices with processing times in days. Please draw a random sample of 60 invoices, calculate the sample mean processing time and sample standard deviation, compute the standard error of the mean, check whether the finite population correction applies, and tell me whether the sample is large enough for the CLT to apply.

OUTPUT –

1. Calculated Statistics

- **Sample Mean ($\bar{x}$):** $\bar{x} = \mathbf{5.2017}$ days
- **Sample Standard Deviation (s):** $s = \mathbf{2.0491}$ days

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2. Finite Population Correction (FPC)

- **FPC Check:** Since the sample represents more than 5% of the total population: $\frac{n}{N} = \frac{60}{850} \approx \mathbf{0.0706}$ $(7.06\% > 5\%)$
- **FPC Flag: APPLIED** (The FPC factor should be used).
- **FPC Factor:** FPC $= \sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{850-60}{849}} \approx \mathbf{0.9646}$

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3. Standard Error of the Mean ($SE(\bar{x})$)

- **Calculation (with FPC):** $SE(\bar{x}) = \text{FPC} \times \frac{s}{\sqrt{n}} = 0.9646 \times \frac{2.0491}{\sqrt{60}} \approx \mathbf{0.2552}$ days *(Note: Without the FPC adjustment, the standard error would be* ≈ 0.2645 *days)*

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4. Central Limit Theorem (CLT) Normality Justification

- **CLT Flag: MET**.
- **Reasoning:** Since the sample size is $n = 60 \geq 30$, the sample size is large enough to justify treating the sampling distribution of the sample mean ($\bar{x}$) as approximately normal under the Central Limit Theorem, regardless of the shape of the underlying population distribution.

1. Does the sample mean support or contradict the vendor's claim of 3 days? By how many SEs does your x̄ differ from 3?

Difference: 5.2017 − 3 = 2.2017 days
In SE units: 2.2017 / 0.2552 ≈ **8.6 SEs**

That's a huge gap — under a normal sampling distribution, being 8+ SEs away from a claimed value is essentially impossible if the claim were true. The sample **strongly contradicts** the vendor's claim of a 3-day average processing time.

2. How precise is your estimate? If SE = 0.4 days, the true mean is likely within ±0.8 days of x̄. Is this precision acceptable for a procurement decision?

With your actual SE of 0.2552 (not the hypothetical 0.4 in the prompt), a rough ~95% interval is x̄ ± 2×SE ≈ 5.20 ± 0.51 days, i.e., roughly **4.70 to 5.71 days**. That's a tight band relative to the 2.2-day gap from the vendor's claim, so yes — this precision is more than sufficient to conclude the claim doesn't hold

3. Two-sentence briefing to a procurement manager? Compose a two-sentence briefing using the numbers.

Based on a random sample of 60 invoices (out of 850), the vendor's actual average processing time is approximately 5.20 days — about 2.2 days slower than their claimed 3-day average, a gap roughly 8.6 standard errors wide, meaning this is very unlikely to be due to sampling chance. I'd recommend flagging this discrepancy with the vendor and requesting either revised SLAs or a corrective action plan.

Reflection Questions

What happens to SE if you double n to 120? (Hint: it does not halve.)

SE scales with $1/\sqrt{n}$, not $1/n$. Going from 60 to 120 multiplies SE by roughly $\sqrt{(60/120)} \approx 0.707$ — so you'd cut SE by about **29%, not 50%**. To actually halve SE you'd need roughly 4x the sample size (240), all else equal. (The FPC factor would also shift slightly as n grows relative to N.)

The agent drew a simple random sample. What if processing times vary by invoice type? What sampling design modification would you suggest?

simple random sampling could give a misleading overall average if, say, international invoices take much longer than domestic ones and got over/under-represented by chance. A better design would be **stratified random sampling** — split the population into strata by invoice type, sample proportionally (or by another allocation rule) within each stratum, then combine. This reduces variance and ensures each type is fairly represented.

What does CLT mean practically — what can you now compute that you otherwise could not?

even though you don't know the true shape of the population distribution of processing times, CLT guarantees that for a large enough sample ($n \geq 100$ is the common rule of thumb), the sampling distribution of $\bar{x}$ is approximately normal. That's what lets us compute standard errors, build confidence intervals, and run z/t-tests on the mean — tools that assume normality — without needing the underlying data itself to be normal.